

Calculus I FINAL

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] (final) Let $f(x) = (x^2 + 2)(3x - 1)$. (a) Differentiate using the product rule. (b) Evaluate $f'(3)$. Show both terms before simplifying.

2. [6 pts] Compute the definite integral $\int_0^5 3x \, dx$. Write an antiderivative, substitute both limits, and state the units of accumulated quantity.

3. [8 pts] A rectangle has perimeter 20 units and one side is 7 units. (a) Write its area as a function of the other side. (b) Explain whether the stated dimensions maximize area.

4. [6 pts] Evaluate $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$. Show the factorization before taking the limit.

5. [6 pts] For a differentiable function, distinguish a critical point from a local maximum and give one test that can classify it.

6. [8 pts] (final) Let $f(x) = (x^2 + 4)(3x - 1)$. (a) Differentiate using the product rule. (b) Evaluate $f'(1)$. Show both terms before simplifying.

7. [6 pts] Compute the definite integral $\int_0^2 3x \, dx$. Write an antiderivative, substitute both limits, and state the units of accumulated quantity.

8. [8 pts] A rectangle has perimeter 14 units and one side is 4 units. (a) Write its area as a function of the other side. (b) Explain whether the stated dimensions maximize area.

9. [6 pts] Evaluate $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$. Show the factorization before taking the limit.

10. [6 pts] For a differentiable function, distinguish a critical point from a local maximum and give one test that can classify it.
